



COLLEGE OF COMPUTING AND INFORMATION TECHNOLOGY – SOUTH VALLEY CAMPUS

Department	: Computer Science	Course Code	: CAI3105/CS460
Lecturer	: Prof. Nashwa El-Bendary	Marks	: 20 Marks
Course	: Deep Learning	Submission	
Submission	Friday	Platform:	Moodle LMS
Deadline:	May 8 th 2026 (11:55PM)		

12th Week project

Project Objective:

This project evaluates the performance of **End-to-End Deep Learning (DL)** classification against **DL-based Feature Learning**. The project requires a comparative study between a fully integrated pre-trained DL model and a hybrid pipeline that utilizes the same pre-trained model for feature extraction followed by a traditional Machine Learning (ML) classifier.

Team Structure and Submission Instructions: This project is designed for collaborative work.

- **Maximum Team Size:** Each project team must **not exceed Four (4) students**.
- **Submission:** Each member of the team must submit one complete report/documentation as a PDF file on Moodle LMS.
- **Documentation & Reporting:** Each team member is required to upload a comprehensive project report in **PDF format** to Moodle LMS. While the core research is a collaborative effort, individual submissions are mandatory for grading purposes. The report must be formally structured according to the specified **Requirements 1, 2, 3, and 4**, listed below, to ensure all technical and analytical components are thoroughly addressed.
- **Code Repository:** Each team member must submit, via Moodle LMS, a link of the **public GitHub repository** containing the complete, runnable implementation. The repository must include all scripts for data preprocessing, Deep Learning (DL) feature extraction, and Machine Learning (ML) classification.
- **Individual Video Presentation:** Each team member must independently record a concise video (approximately **one minute** in duration), upload it to their personal YouTube channel, and submit the link via Moodle LMS. The presentation must cover:
 - **Introduction:** A brief overview of the selected image dataset and the specific pre-trained CNN architecture utilized.
 - **Comparative Analysis:** A summary of the primary results comparing the **End-to-End DL model** against the **ML classifier** powered by learned features.
 - **Technical Insights:** The student's personal evaluation regarding the effectiveness of utilizing pre-trained models as fixed feature extractors versus fully integrated end-to-end training.
- **In-Class Presentation & Discussion:**
 - **Date & Time:** Final presentations will be held in class on **Saturday, May 9th, 2026**, during the scheduled lecture time (The presentation duration for each team is **7–10 minutes**).
 - **(1 mark) Dress Code:** Students must strictly adhere to a **formal or semi-formal dress code**.
 - **(1 mark) A3 Poster:** Each team must generate a **printed A3 poster** according to the provided template to be used during the presentation.
 - **Technical Readiness:** Teams must have a laptop ready with the **dataset** and **runnable code** prepared for technical discussion.

Project Requirements:

➤ Requirement 1: Dataset Selection and Technical Specifications (5 marks)

Teams must select **one** publicly available image dataset from the following suggested benchmarks:

Dataset	Problem Domain	Access Link
Chest X-Ray (Pneumonia)	Medical Diagnosis	Kaggle Dataset / Mendeley Data
Plant Village (Leaf Disease)	Agriculture	Kaggle Dataset / UCI Repository
CIFAR-10	General Object Recognition	Kaggle Dataset / TensorFlow Datasets
Brain MRI (Tumor Detection)	Medical Imaging	Kaggle Dataset / Mendeley Data

For the selected dataset, provide the following:

- 1) (1 mark) **Dataset Metadata:** Source, problem domain, and total number of samples (N).
- 2) (1 mark) **Technical Specifications:** Describe the image resolution (e.g., 224 x 224), color channels (RGB or Grayscale), and the total number of classes.
- 3) (1 mark) **Data Preprocessing:** Detail the steps taken to prepare the data, including resizing and pixel normalization.
- 4) (1 mark) **Data Augmentation:** Describe any Data Augmentation techniques employed (e.g., rotation, flipping, cropping, .. etc.) to improve model robustness and generalize performance. Students must provide a clear justification for the selection of these specific techniques and their necessity for the chosen dataset.
- 5) (1 mark) **Data Splitting:** Specify the utilized partition of data into Training, Validation, and Testing sets (e.g., images split 80/20 or 70/30 Training/Testing sets).

➤ Requirement 2: DL Model Selection (4 marks)

- 1) (1 mark) Select **one** pre-trained Convolutional Neural Network (CNN) architecture (**ResNet50**, **VGG16**, **EfficientNet-B0**, or **MobileNetV1**) to serve as the foundation for both comparative approaches:
 - [Approach-1: The pre-trained model is used as a fixed feature extractor to generate input feature vectors for a traditional ML (**Support Vector Machine (SVM)** or **Logistic Regression**) classifier].
 - [Approach-2 (End-to-End): The same pre-trained model is used as a complete, integrated deep learning classifier].
- 2) (2 marks) Provide a technical justification for the selected pre-trained DL model [You must include a diagram or description of the **model's architecture** sourced from the literature, ensuring the original reference is properly cited].
- 3) (1 mark) Provide a detailed **Table** listing the hyperparameters used for the DL and ML models selected for both implementations (e.g., learning rate, batch size, number of epochs, optimizer, and SVM kernel type).

➤ **Requirement 3: Implementation Framework (5 marks)**

- **(3 marks) Approach-1: DL-Based Feature Learning + ML Classifier**

- 1) Utilize the selected pre-trained model as a feature extractor to convert the images into numerical feature vectors.
- 2) Input these learned features into a traditional ML algorithm: **Support Vector Machine (SVM)** with a linear kernel or **Logistic Regression (LR)**.
- 3) Report performance measures: **Accuracy, Precision, Recall, F-measure, and Confusion Matrix**.

- **(2 marks) Approach-2: End-to-End Deep Learning Model**

- 1) Implement the same selected pre-trained CNN architecture as a complete end-to-end classification model.
- 2) Report performance measures: **Accuracy, Precision, Recall, F-measure, and Confusion Matrix**.

➤ **Requirement 4: Comparative Analysis and Insights (4 marks)**

- 1) **(2 marks)** Conduct a comparative analysis presenting the results from **Approach-1** (ML classifier using DL-based learned features) and **Approach-2** (End-to-End DL model). This comparison must be supported by the reported performance measures for both implementations [It is highly recommended to utilize infographics, such as bar charts for metric comparisons or learning curves (loss/accuracy), to clearly illustrate the comparative analysis between the two approaches].
- 2) **(2 marks)** Write a concise conclusion discussing the experimental results by addressing the following guidance questions:
 - i. How did the performance of the traditional ML classifier (based on DL-based learned features) compare to the performance of the end-to-end deep learning model?
 - ii. What are the observed advantages or limitations of using pre-trained DL models as fixed feature extractors for traditional ML algorithms versus allowing the model to learn and classify in a single integrated pipeline?
 - iii. Which approach proved to be more efficient in terms of training time?
 - iv. Based on your findings, which strategy would you recommend for a resource-constrained environment (e.g., mobile or edge computing) versus a high-performance environment?

Bonus Marks (+2 Marks)

- Students may earn up to 2 bonus marks in addition to the original 20-mark total for the CAI3105-CS460-DL-12th week project by completing the following:
- Extra CNN Model **(1 Mark)**: Apply an additional CNN architecture using the same dataset.
 - Extra Dataset **(1 Mark)**: Evaluate the primary Deep Learning model on a second dataset.

In both cases, a comparative analysis of the performance results must be presented in the final submission.